

Vectors: the Geometry of Data

Data becomes arrows; similarity becomes an angle

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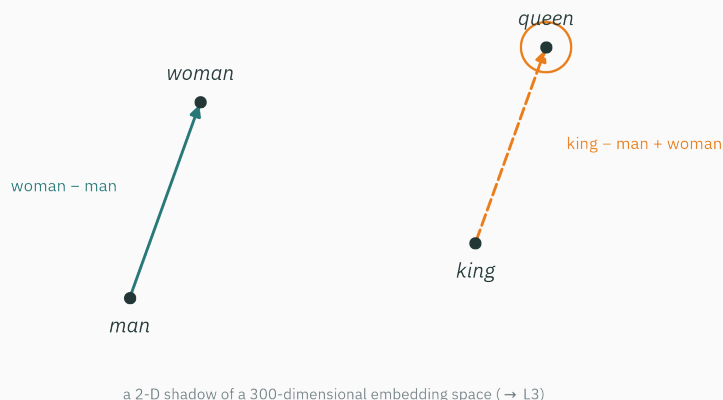
Mathematical Foundations for AI · IIT Gandhinagar

Word embeddings as vectors

The word-embedding example from Lecture 1

Lecture 2 explained the `nan` loss. The learning-rate example returns in L17. We now study the vector relation:

$$\text{king} - \text{man} + \text{woman} \approx \text{queen}$$



The relation is a statement about vector displacement and direction; we will verify both numerically.

Exhibit B · “people like you”

Every streaming app claims to know “people like you”. But taste isn’t a number — what could “like you” even **mean**, mathematically?

Film	You	Priya	Rohan
Dangal	5	4	0
3 Idiots	4	5	1
Interstellar	1	0	5
Chak De! India	?	5	2

Who is more “like you” — Priya or Rohan? **By how much?** And should the app pitch you *Chak De!*? Within the hour you’ll compute all three answers — with the same formula that explains king – man + woman.

Today's one idea

A data point is a **vector**. Similarity between data points is **geometry** — ultimately, an angle.

The route, tool by tool:

Step	Tool	Use
length	norms $\ x\ $	how big, how far apart
angle	dot product, cosine	both questions above
shadow	projection	the germ of least squares & PCA
grid	span, basis	what coordinates even mean

Module 1 starts here: L3 vectors $\rightarrow$ L4 matrices (they **move** vectors) $\rightarrow$ L5 eigenvectors $\rightarrow$ L6 SVD & PCA.

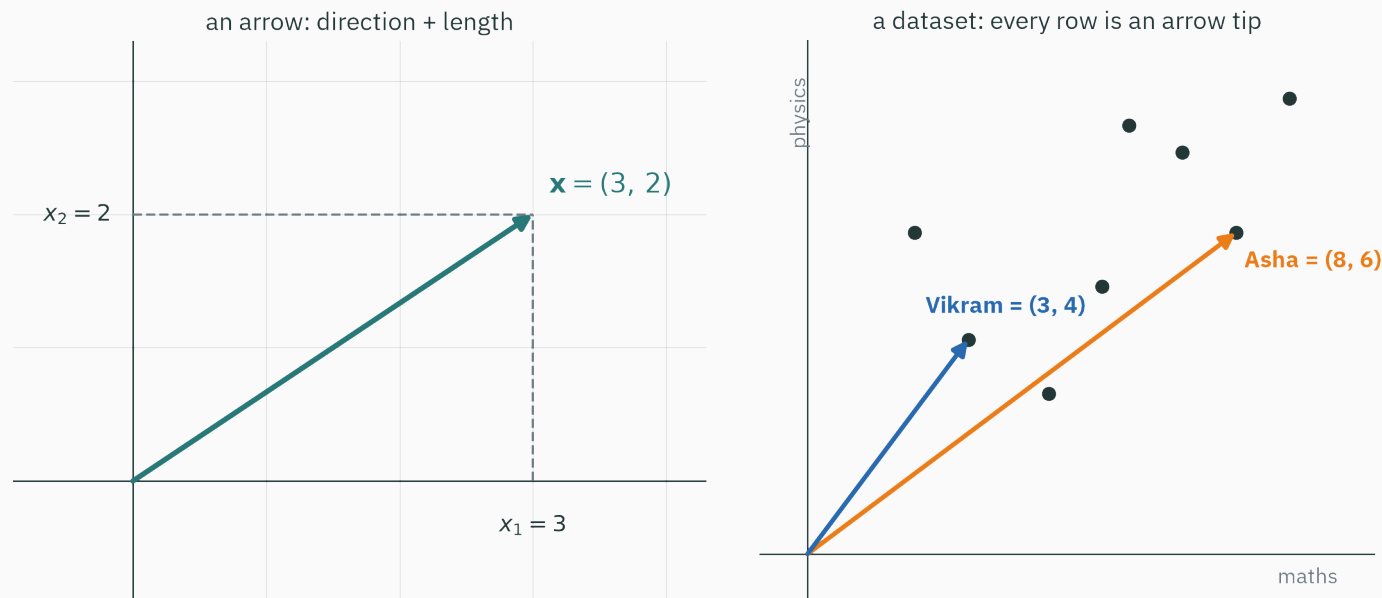
Learning outcomes

By the end of this lecture you will be able to:

1. Read a vector both ways: an **arrow** in space and a **row** of a dataset.
2. Measure size with **L1 / L2 norms** — and draw each norm’s “unit ball”.
3. Compute the **dot product** three ways: multiply–add, lengths $\times \cos \theta$, shadow.
4. Derive $\cos \theta = x^\top y / (\|x\| \|y\|)$ from the school law of cosines (★).
5. Rank **real word embeddings** by cosine similarity and evaluate the opening analogy.
6. **Project** one vector onto another and verify the leftover is orthogonal.
7. Say when vectors **span** a line vs the plane, and what an (orthonormal) **basis** buys.

Vectors: arrows and data rows

One object, two readings



Left: $(3, 2)$ is an **arrow** — a direction plus a length. Right: the same kind of pair is a **row of data** — Asha scored $(8, 6)$ in (maths, physics).

One object, two readings — an arrow you can draw, a row you can store. All of today is about switching between them at will.

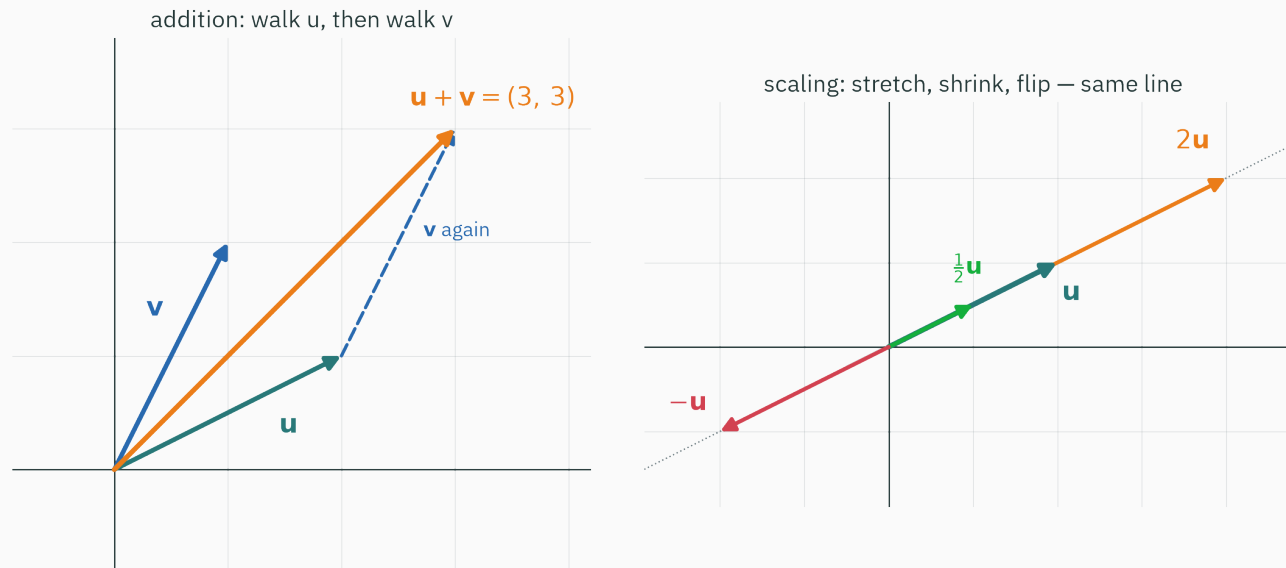
Everything you meet in AI is one of these

Object	The vector	Lives in
Asha's marks	$(8, 6)$	$\mathbb{R}^2$
your film ratings	$(5, 4, 1, \dots)$	$\mathbb{R}^{\# \text{films}}$
the word “king” (GloVe)	50 learned numbers	$\mathbb{R}^{50}$
one MNIST digit	$28 \times 28 = 784$ pixel values	$\mathbb{R}^{784}$
3 s of speech at 16 kHz	48,000 samples	$\mathbb{R}^{48000}$

L2 callback: every entry is a float32 — each coordinate lives on last lecture's unevenly spaced number line.

From here on, “data” means: n points in $\mathbb{R}^d$ — n = how many, d = how rich.

Two moves only: add, and scale



Addition: walk u , then walk v — tip to tail. Scaling: stretch, shrink, or flip along the same line.

These two moves are the whole meaning of “linear”. Everything in Module 1 — combinations, span, matrices — is built from exactly these.

The moves, in numbers and code

With $u = (2, 1)$, $v = (1, 2)$: everything happens slot by slot —

$$u + v = (3, 3) \quad 2u = (4, 2) \quad -u = (-2, -1)$$

```
u, v = np.array([2., 1.]), np.array([1., 2.])
u + v      # array([3., 3.])
2 * u      # array([4., 2.])
u * v      # array([2., 2.]) ← slot-by-slot, NOT the dot product
```

$u * v$ multiplies slot-by-slot and keeps a **vector**. The product that carries geometry — $u @ v$ — collapses two vectors into one number. It gets its own section, two sections from now.

The notation contract

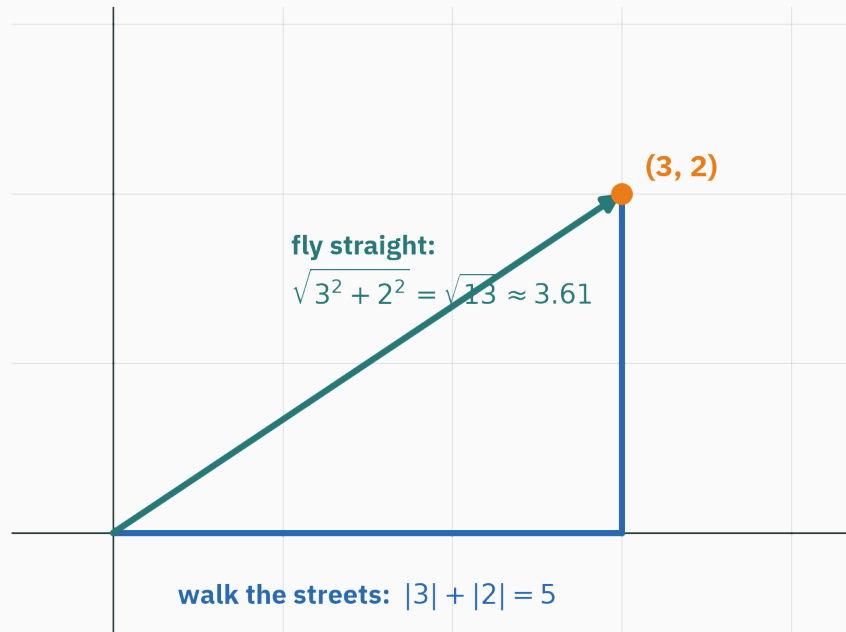
$$\mathbf{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_d \end{pmatrix} \in \mathbb{R}^d \quad \mathbf{x}^\top = (x_1, \dots, x_d)$$

- **bold lowercase** $\mathbf{x}$: vectors (columns by default) · *italic* x_i : scalars & components
- CAPS A, X : matrices — from L4 on; the dataset is $X \in \mathbb{R}^{n \times d}$, one point per row
- n data points, d dimensions; an unadorned $\|\mathbf{x}\|$ always means the L2 norm

This is the same contract as MML and the CS229 linear-algebra handout — decks, tutorials, and assignments all speak it. Learn it once, read everything.

Norms: how big is a vector?

Two honest ways to measure one trip



Walk the streets: $|3| + |2| = 5$ blocks. Fly straight: $\sqrt{3^2 + 2^2} = \sqrt{13} \approx 3.61$.

Both are legitimate “sizes” of the same vector — they answer different questions. Let’s formalize both.

The L2 norm · as the crow flies

$$\|x\|_2 = \sqrt{x_1^2 + x_2^2 + \cdots + x_d^2} = \sqrt{x^\top x}$$

For the trip: $\|(3, 2)\|_2 = \sqrt{13} \approx 3.61$. Works in any dimension — for $(1, -2, 2) \in \mathbb{R}^3$:
 $\sqrt{1 + 4 + 4} = 3$.

- This is **the** default: $\|x\|$ with no subscript means L2.
- It is Pythagoras, applied $d - 1$ times — flight distance in $\mathbb{R}^d$.

Keep the inner form $\sqrt{x^\top x}$ in view: **the length is secretly a product of x with itself** — that observation runs the next section.

The L1 norm · street by street

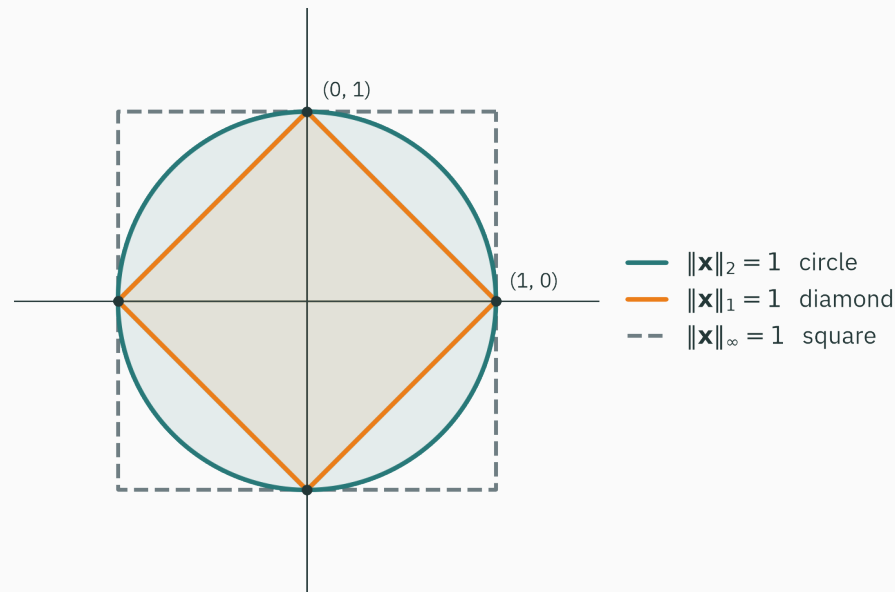
$$\|x\|_1 = |x_1| + |x_2| + \cdots + |x_d|$$

Worked example — a model's errors on four students, $e = (2, -1, 3, -4)$:

Norm	Value	What it judges
$\ e\ _1$	$2 + 1 + 3 + 4 = 10$	total miss — every error counts equally
$\ e\ _2$	$\sqrt{30} \approx 5.48$	squares amplify — big misses dominate
$\ e\ _\infty$	$\max = 4$	only the single worst case

One model, three verdicts. **Choosing a norm is choosing what you care about.**

Every norm draws a different “circle”



All three earn the name “norm” by the same three axioms (MML 3.1): positive unless $x = 0$, $\|cx\| = |c| \|x\|$, triangle inequality.

And the L1 diamond has **corners** on the axes — the seed of why L1 regularization zeroes weights out. (ES 335 finishes that story.)

Unit vectors · direction, distilled

$$\hat{x} = \frac{x}{\|x\|} \quad \text{same direction, length exactly 1}$$

Numbers: $x = (3, 2)$: $\hat{x} = (3, 2)/\sqrt{13} \approx (0.832, 0.555)$ — check: $0.832^2 + 0.555^2 \approx 1$ ✓

Every nonzero vector splits cleanly: $x = \|x\| \cdot \hat{x}$ — **how much** × **which way**.

Hold on to this: today's similarity story turns out to be entirely about the **which way** part.

Distance · a norm in disguise

$$d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|$$

Asha (8, 6) vs Vikram (3, 4): difference (5, 2) → flight distance $\sqrt{29} \approx 5.39$, street distance 7.

```
asha, vikram = np.array([8., 6.]), np.array([3., 4.])  
np.linalg.norm(asha - vikram)      # 5.385... (L2)  
np.linalg.norm(asha - vikram, 1)    # 7.0      (L1)
```

Every “nearest” in ML — nearest neighbour, nearest cluster centre, nearest embedding — is this one line of code plus a choice of norm.

One dot product, three readings

Reading 1 · multiply matching slots, then add

$$\mathbf{x}^\top \mathbf{y} = x_1 y_1 + x_2 y_2 + \cdots + x_d y_d = \sum_{i=1}^d x_i y_i$$

The running pair for the rest of this lecture: $\mathbf{x} = (3, 2)$, $\mathbf{y} = (4, 2)$:

$$\mathbf{x}^\top \mathbf{y} = 3 \cdot 4 + 2 \cdot 2 = 16$$

Two vectors in, **one number** out. But what does “16” mean? Hold that thought.

One meaning is already ours — pair $\mathbf{x}$ with itself: $\mathbf{x}^\top \mathbf{x} = 9 + 4 = 13 = \|\mathbf{x}\|^2$.

$\|\mathbf{x}\| = \sqrt{\mathbf{x}^\top \mathbf{x}}$ — the length was a dot product all along.

Reading 2 · lengths × alignment

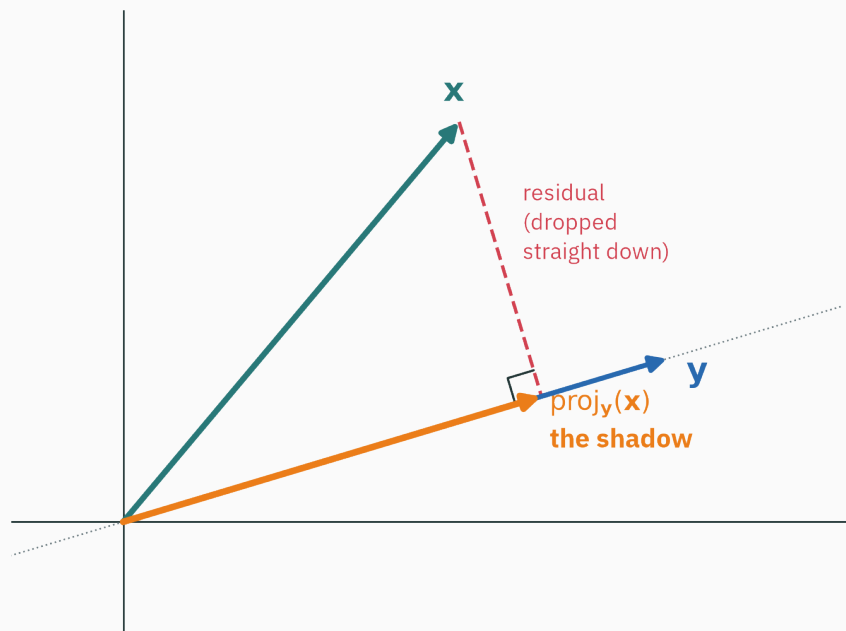
The claim (to be **earned** next section — it is today's ★):

$$x^\top y = \|x\| \|y\| \cos \theta$$

Check it against the running pair: $\|x\| \|y\| = \sqrt{13}\sqrt{20} \approx 16.12$, so

$$\cos \theta = \frac{16}{16.12} \approx 0.992 \quad \Rightarrow \quad \theta \approx 7.1^\circ \quad \text{— two nearly parallel arrows.}$$

Pause on how strange this is: pure multiply–add on coordinates apparently **knows the angle**. No protractor was consulted. We will demand proof shortly.



Dot x against a **unit** direction: $x^\top \hat{y} = \|x\| \cos \theta =$ the signed **length of x 's shadow** on y 's line. Running pair: $16/\sqrt{20} \approx 3.58$.

Read it as: “how much of x points the y way.”

The alignment detector

Angle	$\cos \theta$	$x^\top y$	Interpretation
acute ($\theta < 90^\circ$)	+	+	the vectors agree
right ($\theta = 90^\circ$)	0	0	orthogonal — $x \perp y$, nothing shared
obtuse ($\theta > 90^\circ$)	—	—	they oppose

Check: $(3, 2)^\top (-2, 3) = -6 + 6 = 0 \rightarrow$ perpendicular. The axes too: $e_1^\top e_2 = 0$.

Orthogonal directions carry **no information about each other** — the dot product literally sees nothing of one in the other. This idea returns twice today: residuals, and the nicest bases.

Three readings, one number

```
x, y = np.array([3., 2.]), np.array([4., 2.])
x @ y                                # 16.0      reading 1: multiply-add
nx, ny = np.linalg.norm(x), np.linalg.norm(y)
theta = np.arccos(x @ y / (nx * ny))
np.degrees(theta)                    # 7.125      the angle between them
nx * np.cos(theta)                   # 3.578      reading 3: shadow length
(x @ y) / ny                         # 3.578      same shadow, no trig needed
```

Plant for L4: a matrix multiply is nothing but these, en masse — every entry of Wx is one dot product $w_i^\top x$, one “how much of x lies along my direction?” question.

$x = (3, 2)$. Which y is **orthogonal** to x ?

A. $y = (2, 3)$

B. $y = (-2, 3)$

C. $y = (3, -2)$

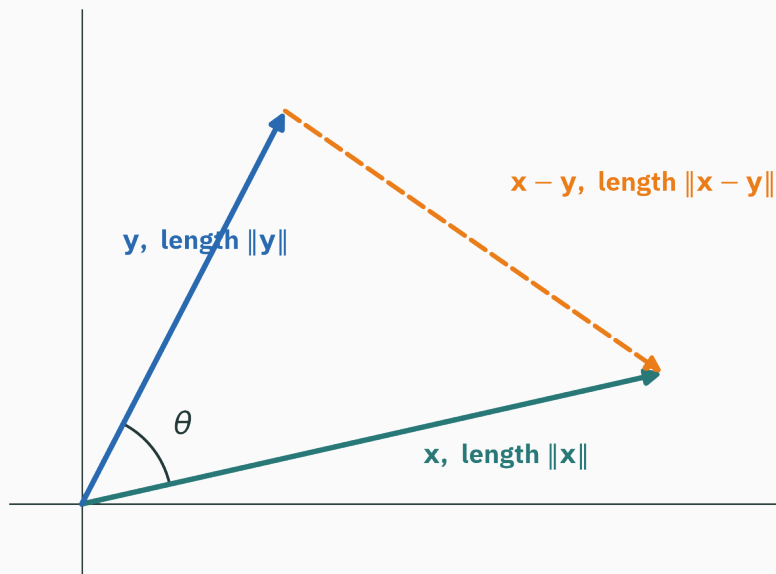
D. $y = (-3, -2)$

Correct: B — $(-2, 3)$, since $3 \cdot (-2) + 2 \cdot 3 = 0$

Why: in 2-D, “flip the slots, negate one”: $(-x_2, x_1)$ is always $\perp (x_1, x_2)$. A gives 12 and C gives 5 (acute); D gives -13 — **opposing**, but only exactly zero means perpendicular.

Why multiply–add measures angle

Claim (★): for any two nonzero vectors, $x^\top y = \|x\| \|y\| \cos \theta$.



The two sides of the claim speak different languages — coordinates vs geometry. The bridge: compute the third side $\|x - y\|^2$ **twice**, once in each language, and compare.

Step 1 · algebra expands the third side

D DERIVATION

$$\|x - y\|^2 = (x - y)^\top (x - y) \quad (\text{a norm}^2 \text{ is a self-dot})$$

$$= x^\top x - x^\top y - y^\top x + y^\top y \quad (\text{expand the brackets})$$

$$= \|x\|^2 + \|y\|^2 - 2x^\top y \quad (\text{since } y^\top x = x^\top y)$$

Only two facts were used — brackets distribute, and order doesn't matter — and both read straight off the multiply-add formula: check $x^\top (y + z) = \sum_i x_i (y_i + z_i) = x^\top y + x^\top z$.

The **law of cosines** (Class 10): any triangle with sides a, b, c and angle γ opposite c obeys

$$c^2 = a^2 + b^2 - 2ab \cos \gamma \quad (\text{Pythagoras, plus a tilt correction})$$

Our triangle has sides $a = \|x\|$, $b = \|y\|$, $c = \|x - y\|$, and $\gamma = \theta$:

$$\|x - y\|^2 = \|x\|^2 + \|y\|^2 - 2\|x\|\|y\| \cos \theta$$

Sanity check the correction: at $\theta = 90^\circ$ the cosine dies and pure Pythagoras remains. ✓

Both computations measured the **same number** $\|x - y\|^2$:

$$\|x\|^2 + \|y\|^2 - 2x^\top y = \|x\|^2 + \|y\|^2 - 2\|x\|\|y\|\cos\theta$$

Cancel the squared lengths, divide by -2 :

$$x^\top y = \|x\|\|y\|\cos\theta \iff \cos\theta = \frac{x^\top y}{\|x\|\|y\|}$$

Check on the running pair: $16 = 16.12 \times 0.992 \checkmark$ ($\theta \approx 7.1^\circ$, as promised).

Free corollary: $|\cos\theta| \leq 1$ forces $|x^\top y| \leq \|x\|\|y\|$ — that's **Cauchy–Schwarz**, bought for nothing.

**Cosine similarity: geometry
becomes meaning**

Angles in $\mathbb{R}^{300}$, by decree

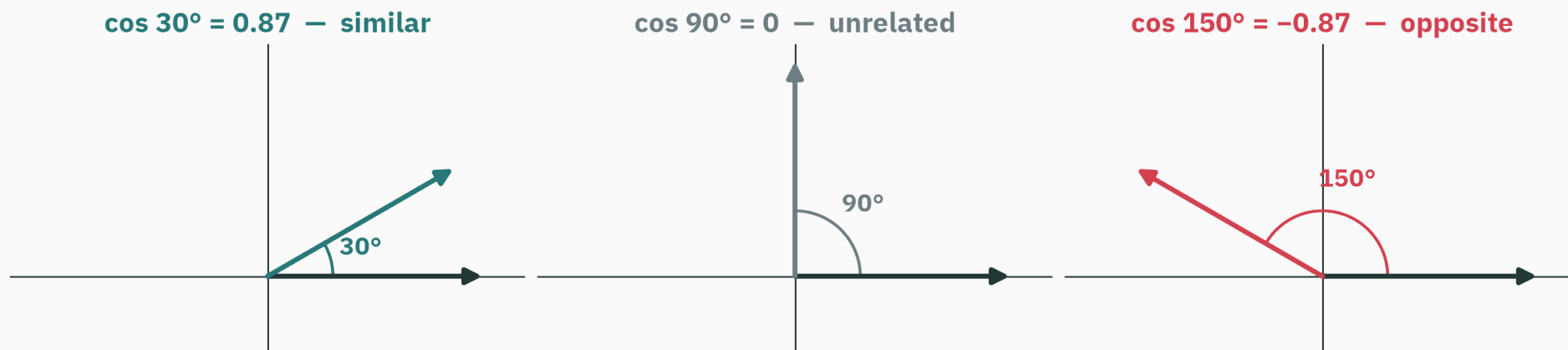
In $\mathbb{R}^2$ we **saw** θ and proved the formula. In $\mathbb{R}^{300}$ nobody sees anything — so flip the logic and let the formula **define** the angle:

$$\cos_{\text{sim}}(\mathbf{x}, \mathbf{y}) := \frac{\mathbf{x}^\top \mathbf{y}}{\|\mathbf{x}\| \|\mathbf{y}\|} = \hat{\mathbf{x}}^\top \hat{\mathbf{y}} \quad \text{— a dot product of unit vectors.}$$

This is legal and safe: Cauchy–Schwarz keeps it inside $[-1, 1]$, and any two vectors span an ordinary 2-D plane where it **is** the visible angle.

Cosine is scale-invariant: $\cos_{\text{sim}}(10\mathbf{x}, \mathbf{y}) = \cos_{\text{sim}}(\mathbf{x}, \mathbf{y})$. **Lengths cancel — only **directions** are compared.**

Cosine similarity at three angles



+1: same story · 0: nothing shared · -1: opposite story. For data, read: **similar / unrelated / opposed**.

“People like you”, by hand

Ratings on the three films everyone has seen: you = (5, 4, 1), Priya = (4, 5, 0), Rohan = (0, 1, 5).

$$\text{you} \cdot \text{Priya} : \frac{5 \cdot 4 + 4 \cdot 5 + 1 \cdot 0}{\sqrt{42}\sqrt{41}} = \frac{40}{41.5} \approx 0.96 \Rightarrow \theta \approx 15^\circ$$

$$\text{you} \cdot \text{Rohan} : \frac{0 + 4 + 5}{\sqrt{42}\sqrt{26}} = \frac{9}{33.0} \approx 0.27 \Rightarrow \theta \approx 74^\circ$$

“People like you” = vectors at a small angle to yours. Priya’s 5★ for *Chak De!* becomes your recommendation. Exhibit B: closed.

A word is 50 floats

GloVe (2014): read 6 billion words of text; assign every word a vector so that words appearing in **similar contexts** get similar vectors. No dictionary, no grammar — just co-occurrence.

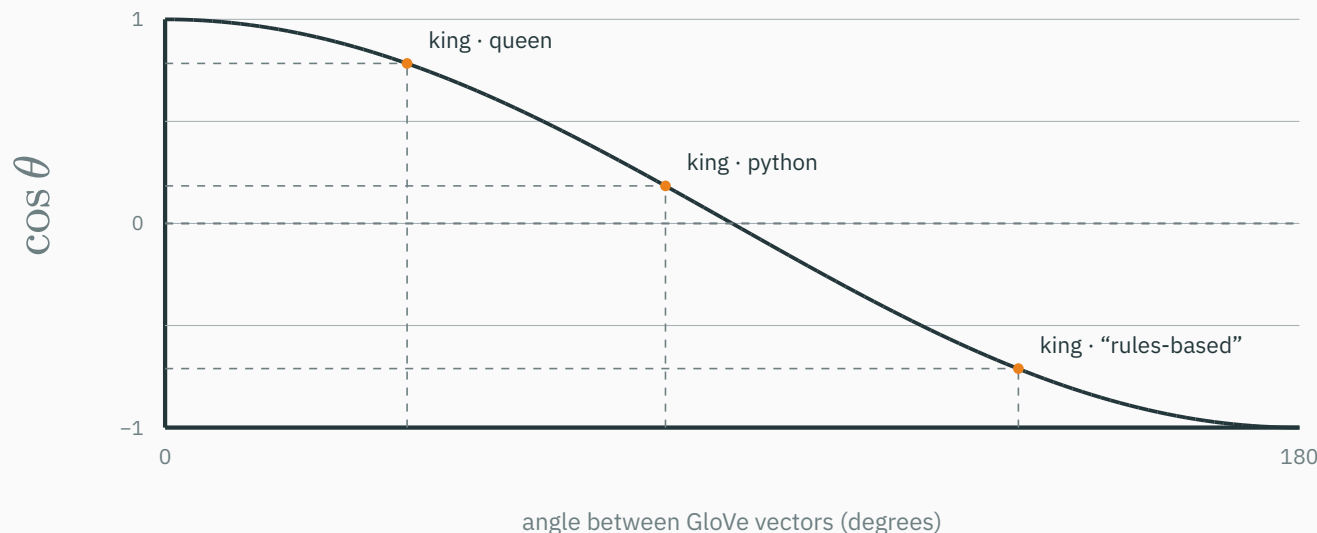
```
import gensim.downloader as api
glove = api.load("glove-wiki-gigaword-50")  # 400k words × 50 floats
glove["king"][:6]
# array([ 0.5045,  0.6861, -0.5952, -0.0228,  0.6005, -0.135 ])
```



```
def cos_sim(u, v):
    return u @ v / (np.linalg.norm(u) * np.linalg.norm(v))
cos_sim(glove["king"], glove["queen"])  # 0.784 →  $\theta \approx 38.4^\circ$ 
```

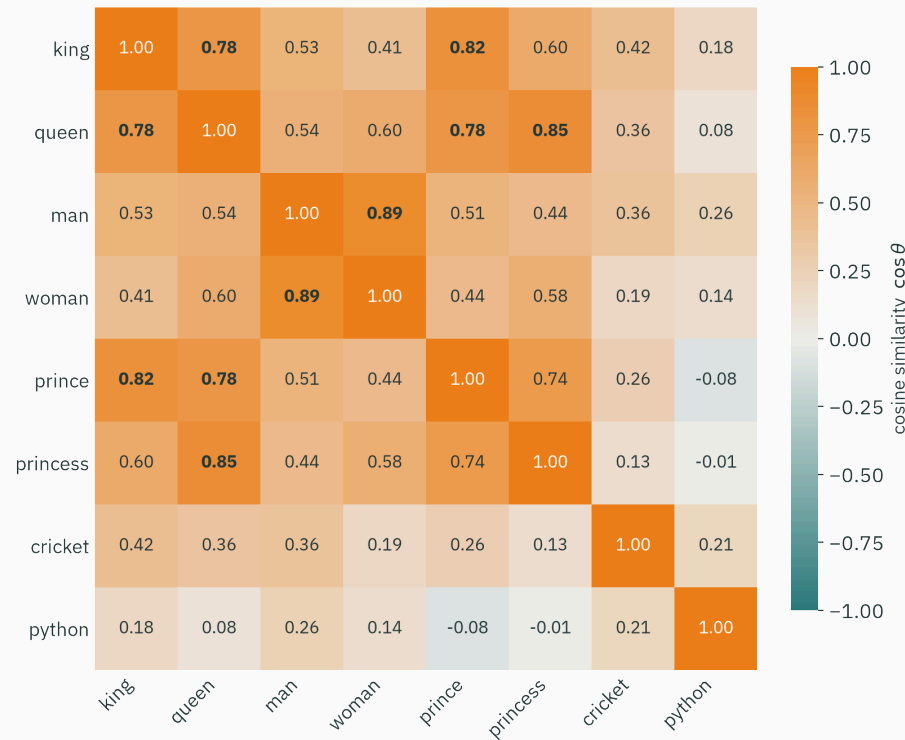
Coordinate 17 of “king” means... nothing anyone can name. Individually the 50 float32s (L2!) are gibberish — **the geometry between words carries the meaning.**

Cosine similarity for real word embeddings



More readings: uncle · aunt lands at 40.3°, king · cricket at 65.1°. Semantic neighbours sit at small angles; strangers drift toward 90° and beyond.

All 28 pairs at once



Blocks are the point: a royal cluster (king–prince 0.82, queen–princess 0.85), a tight man–woman pair (0.89) — and python cold to all, even **opposed** to prince (−0.08).

Evaluate the word analogy

The bet behind the arithmetic:

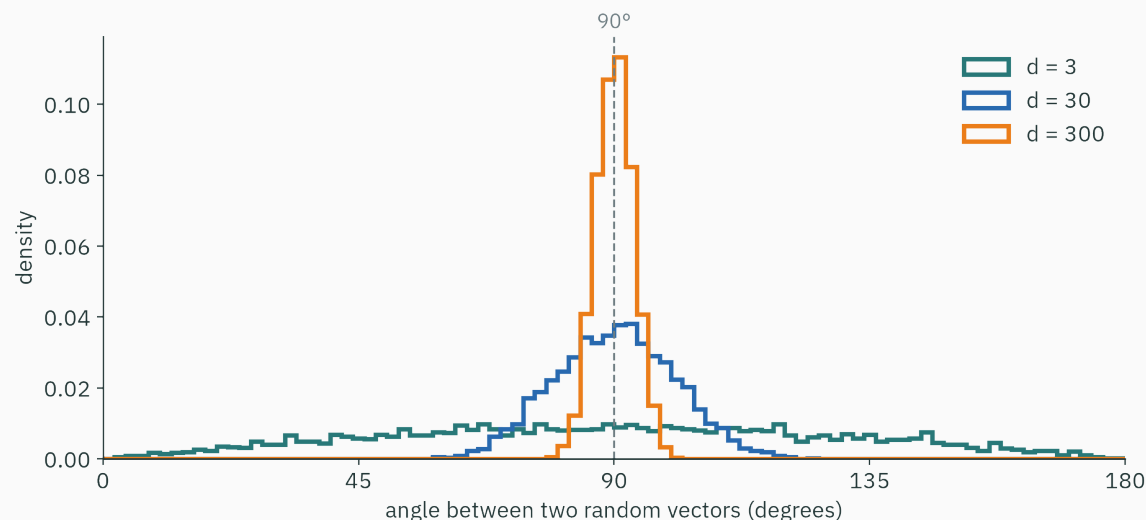
$$v_{\text{king}} - v_{\text{man}} + v_{\text{woman}} = v_{\text{king}} + \underbrace{(v_{\text{woman}} - v_{\text{man}})}_{\text{a learned gender shift}}$$

```
glove.most_similar(positive=["king", "woman"], negative=["man"])[0]  
# ('queen', 0.852) ← nearest of all 400k words, by cosine
```

The subtraction never “understood” gender. **Directions in this space consistently encode relationships** — man→woman and king→queen are nearly parallel arrows.

Honest small print: `most_similar` excludes the query words; keep them in and **king itself** outranks queen. The parallelogram is real but approximate — hence “ $\approx$ ”, never “ $=$ ”.

In high dimensions, strangers are orthogonal



Random pairs in $\mathbb{R}^3$: any angle goes. In $\mathbb{R}^{300}$: everything piles up at 90° — **unrelated is the default state.**

So king · queen at 38° screams against that background. And the pile-up is good news: high- d space fits astronomically many nearly-perpendicular directions — room for 400k words.

Projections: shadows and residuals

From shadow length to shadow vector

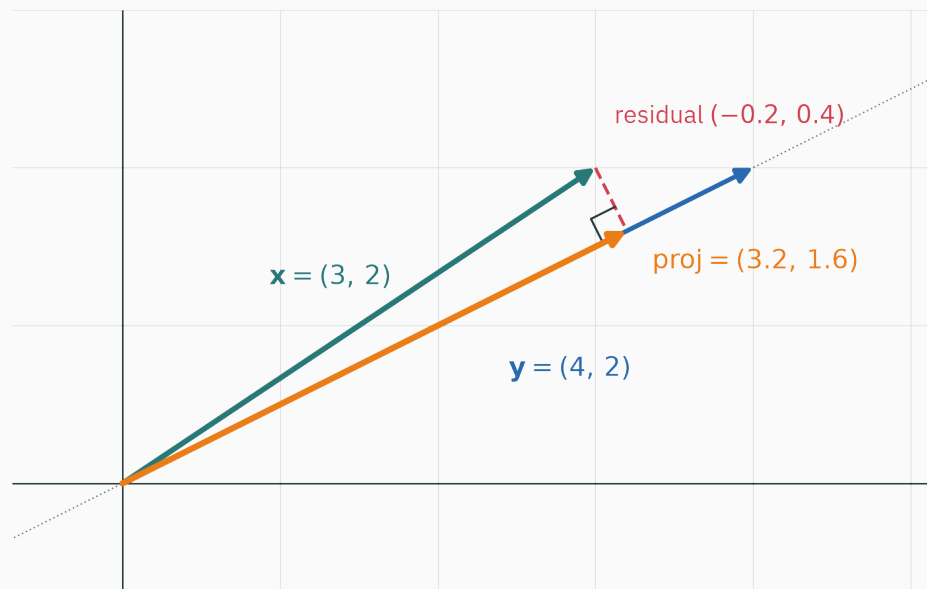
Reading 3 gave a **number** — the shadow's length $x^\top \hat{y}$. The shadow itself is that length, aimed along $\hat{y}$:

$$\text{proj}_{\mathbf{y}}(\mathbf{x}) = (x^\top \hat{\mathbf{y}}) \hat{\mathbf{y}} = \frac{x^\top \mathbf{y}}{\mathbf{y}^\top \mathbf{y}} \mathbf{y}$$

(The right form just substitutes $\hat{\mathbf{y}} = \mathbf{y}/\|\mathbf{y}\|$ twice and uses $\|\mathbf{y}\|^2 = \mathbf{y}^\top \mathbf{y}$.)

Read it: **the multiple of $\mathbf{y}$ that best imitates $\mathbf{x}$** . Whatever's left — the residual $\mathbf{x} - \text{proj}$ — is the part of $\mathbf{x}$ that $\mathbf{y}$ cannot explain.

The shadow, in numbers you can check



$$\frac{x^\top y}{y^\top y} = \frac{16}{20} = 0.8 \Rightarrow \text{proj} = 0.8 (4, 2) = (3.2, 1.6)$$

Residual: $x - \text{proj} = (-0.2, 0.4)$ — and $(-0.2, 0.4)^\top (4, 2) = -0.8 + 0.8 = 0$.

Perpendicular, **exactly**.

Why perpendicular means “closest”

Slide a candidate point along y 's line and connect it to x : the connector is shortest where it meets the line at a right angle — any other landing point makes it a hypotenuse over that leg.

$\text{proj}_y(x)$ is the **best approximation of x on y 's line** — and “best” is certified by an orthogonal residual.

Widen “ y 's line” to a whole plane of allowed directions and this same right-angle picture becomes **least squares** (L16) and **PCA** (L6). A surprising amount of “learning” is projecting.

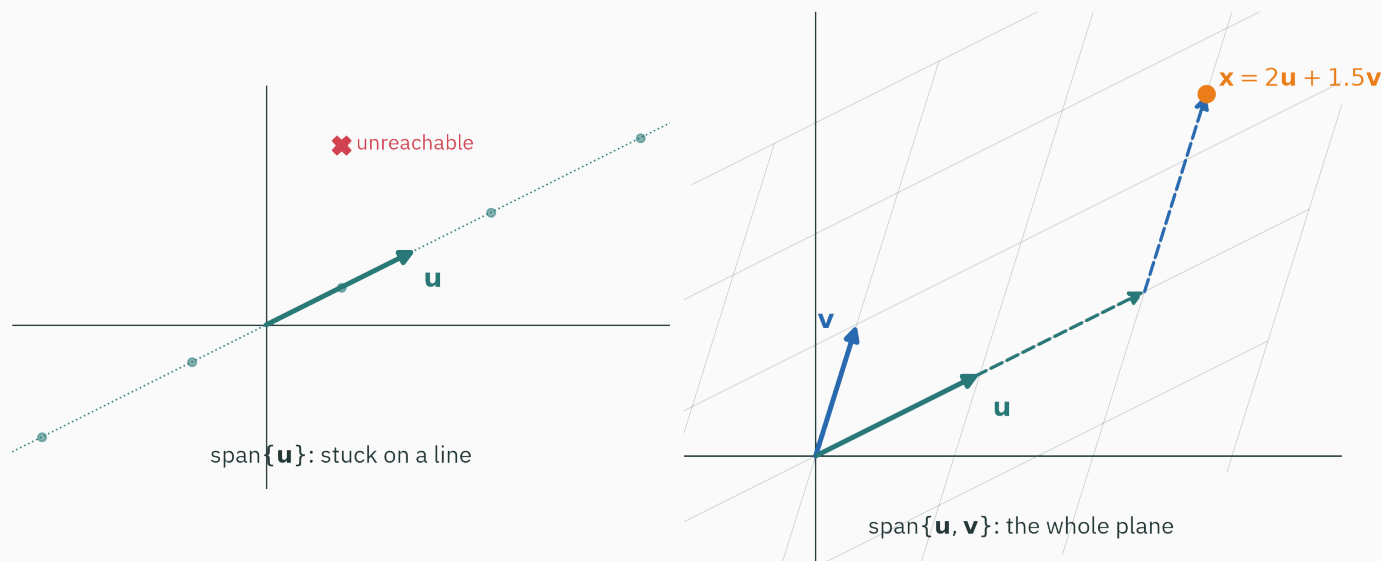
Projection in three lines

```
x, y = np.array([3., 2.]), np.array([4., 2.])
proj = (x @ y) / (y @ y) * y      # array([3.2, 1.6])
r = x - proj                      # array([-0.2,  0.4])
r @ y                             # 0.0 – the residual is  $\perp$ , always
```

The zero is not luck; it is the definition doing its job. You will reuse exactly these lines inside least squares (L16) and PCA (L6).

Basis and span: the grid is a choice

Span · where can your moves take you?



$\text{span}\{u\} = \{\alpha u\}$: scaling alone keeps you on a **line**. $\text{span}\{u, v\} = \{\alpha u + \beta v\}$: with two genuinely different directions, walk u 's then v 's and **every** point of the plane is reachable.

When a second vector adds nothing

Take $u = (2, 1)$ and $v = (4, 2) = 2u$:

$\alpha u + \beta v = (\alpha + 2\beta) u$ — every 'combination' is still a multiple of u .

Linearly independent = no vector in the set is a combination of the others.

Dependent = a redundant direction; the span doesn't grow.

Data version: a feature `total = maths + physics` adds a **column** to your table but **no new direction** to the data. That redundancy is why matrices have rank (L4) and why data compresses (L6).

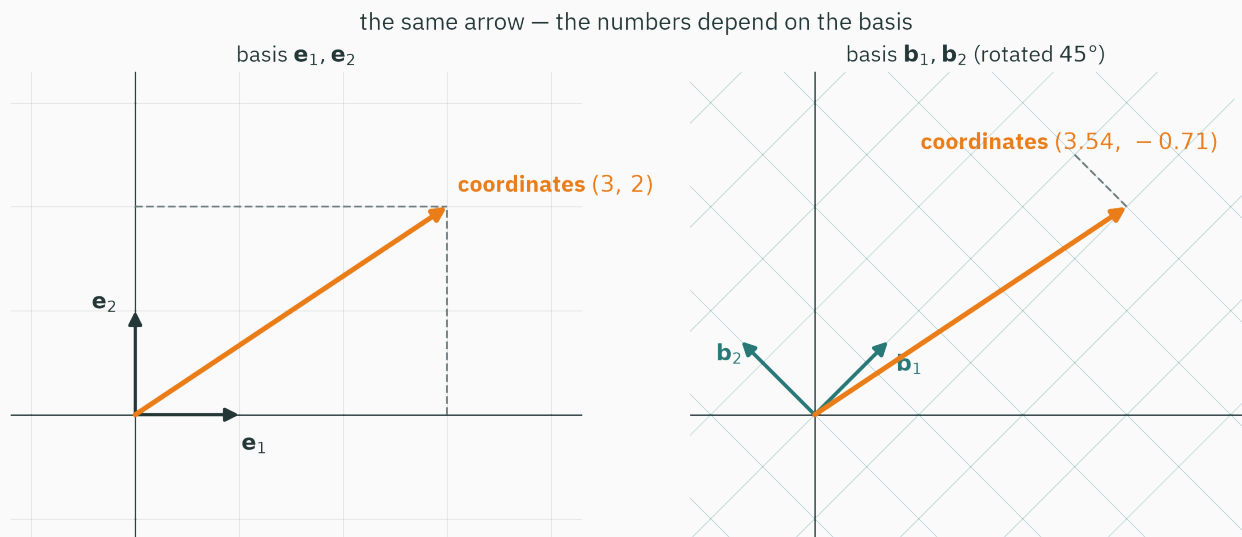
INTERACTIVE

↗ textbooks.math.gatech.edu/ila/spans.html

Interactive Linear Algebra (GaTech) — in the span demos, drag u and v and watch $\text{span}\{u, v\}$ collapse from the whole plane to a line the instant the arrows align.

Two minutes of dragging beats twenty static slides — do it tonight. This free textbook is the reference explorable for all of Module 1 (L3–L6).

A basis is a chosen grid



A **basis** = a set of vectors that is (1) independent and (2) spans the space.
Consequence: every vector gets **exactly one address** — its coordinates.

Same arrow, two grids, two addresses: (3, 2) vs (3.54, -0.71). **The arrow is real; the numbers are an artifact of the grid you chose.**

Orthonormal grids read themselves

If b_1, b_2 are **unit length** and **mutually perpendicular** (orthonormal), addresses come free — each coordinate is a projection:

$$x = (x^\top b_1) b_1 + (x^\top b_2) b_2$$

Check against the figure's rotated grid — $b_1 = (1, 1)/\sqrt{2}$, $b_2 = (-1, 1)/\sqrt{2}$, $x = (3, 2)$:
 $x^\top b_1 = 5/\sqrt{2} \approx 3.54$ and $x^\top b_2 = -1/\sqrt{2} \approx -0.71$ ✓ — the figure's numbers.

In an orthonormal basis, coordinates are projections: reading an address costs d dot products.

L6 teaser: PCA = let the data cloud elect its own favourite orthonormal grid.

Commit to all four before the reveal.

A. $\cos \theta$ between x and y is 0.8. Your teammate rescales $x \rightarrow 10x$. New cosine?

C. $\text{proj}_y(y) = ?$

B. $\text{span}\{(2, 1), (4, 2)\}$: a point, a line, or the whole plane?

D. Can **three** vectors in $\mathbb{R}^2$ be linearly independent?

- A.** 0.8 — unchanged. The 10 scales $x^\top y$ and $\|x\|$ alike and cancels. Cosine sees direction only.
- B.** A line — $(4, 2) = 2 \cdot (2, 1)$: the second move is the first in disguise.
- C.** y itself — the coefficient is $y^\top y / y^\top y = 1$: a vector is its own shadow on its own line.
- D.** No — two independent arrows already span $\mathbb{R}^2$; any third is a combination of them. “How many independent directions fit” is exactly what **dimension** means.

Practice problems

Try on paper; verify in NumPy.

1. For $x = (1, -2, 2)$: compute $\|x\|_1$, $\|x\|_2$, $\|x\|_\infty$. Which is biggest — and is that always so?
2. A new user rates $(1, 1, 5)$. Cosine with you $= (5, 4, 1)$? More or less “like you” than Rohan’s 0.27?
3. Find t so that $(3, 2) \perp (4, t)$.
4. Project $x = (1, 4)$ onto $y = (2, 0)$; write $x = \text{shadow} + \text{residual}$ and verify the residual is $\perp y$.
5. Show $(1, 1) \perp (1, -1)$, normalize both, and write $(3, 2)$ in this basis via two dot products.
6. In the heatmap, $\text{king} \cdot \text{cricket} \approx 0.42$ but $\text{prince} \cdot \text{python} \approx -0.08$. One-sentence story for each number?

The reading map

Source	What it gives you
MML Ch 2.1–2.4	vectors and vector spaces — today's arrows, done carefully
MML Ch 3.1–3.3	norms, inner products, distances — today's rulers
CS229 linear-algebra notes	the course's reference handout; keep it at hand through L6
3b1b <i>Essence of Linear Algebra</i> ch 1–3	today in moving pictures; chapter 3 is the perfect L4 pre-watch

All free: mml-book.github.io · cs229.stanford.edu/section/cs229-linalg.pdf · the 3b1b playlist on YouTube. Quiz questions come from lecture beats, and every beat above has a book home.

Lecture 3 — summary

- **A vector is an arrow AND a data row**; a dataset is n arrows in $\mathbb{R}^d$ (each entry a float32 — L2).
- **Norms measure size**: L2 flight, L1 streets, L^∞ worst case — unit balls: circle, diamond, square.
- **One dot product, three readings**: $\sum_i x_i y_i = \|x\| \|y\| \cos \theta = \text{shadow} \times \text{length}$.
- ★ The **law of cosines** turns multiply–add into an angle meter — and **defines** angles in $\mathbb{R}^{300}$.
- **Cosine similarity** ranks words, films, and people; $\text{king} - \text{man} + \text{woman} \approx \text{queen}$ compares approximately parallel displacement vectors.
- **Projection** = closest point on a line; the residual is orthogonal — the germ of least squares and PCA.
- **Span** is where your moves reach; a **basis** is the grid you narrate coordinates in; orthonormal grids read coordinates by dot products.

Similarity is an angle.

Next: **Matrices as Linear Maps** — the spreadsheet stands up and starts moving: every neural-network layer is a matrix pushing your vectors around.